

Bought, Not Added: Provisional Emission with Retrospective Revision for Streaming 3D Perception

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Abstract

001 Streaming 3D perception must emit objects immediately,
002 yet its consumers need one physical object to stay one ob-
003 ject. Temporal confirmation, which reports a track only af-
004 ter N observations, is the standard remedy, but it either
005 delays those observations or discards them. At matched
006 detection and identity budgets, and against a confidence-
007 accumulation lifecycle, we find that the module’s associa-
008 tion gain is real but is paid for in detection recall, DetA and
009 HOTA. The gain is bought rather than added. A second de-
010 tector narrows it further: it holds in the sensor frame and
011 is zero or negative in the world frame. We therefore pro-
012 pose provisional emission with retrospective revision: ev-
013 ery observation is emitted at its own frame under a provi-
014 sional status, and later evidence confirms, relabels or re-
015 tracts it instead of withholding it. A four-frame retraction
016 horizon exactly reproduces the unbounded provisional out-
017 put. On nuScenes it retains 160,938 observations (+43.6%)
018 that prefix-deleting causal emission discards, at zero emis-
019 sion latency, and revision raises AMOTA by 0.0823 — while
020 giving up DetA, HOTA and IDF1 against that arm. On
021 ScanNet200 the same rule reduces class-label switching
022 by 26.9% at a matched identity count. We offer a differ-
023 ent operating point in the detection–association–latency–
024 revisability trade-off, not a uniformly better tracker.

025 1. Introduction

026 Streaming 3D perception commits online: a map or a plan-
027 ner’s object memory consumes each frame as it arrives [10–
028 12, 16]. Per-frame emission makes that commitment cheap
029 and unstable — a re-detected object arrives as a new iden-
030 tity, an open-vocabulary label changes class between frames
031 — and per-frame detection metrics neither see nor penalise
032 it. Temporal confirmation is the natural correction, and an
033 old one [3, 4, 32, 33]: report a track only once it has been
034 observed N times, so transient errors never reach the con-
035 sumer. But a rule that reports later necessarily reports less;

it either delays commitment or withholds what it is not yet
sure of, and both are paid on the streaming path.

Whether that buys anything is harder to establish than it
looks. Confirmation does not merely reorder a stream — it
changes how many detections enter it and how many iden-
tities the system carries — and detection- and association-
oriented metrics are strongly coupled to both. A confirmed
stream compared against an immediately emitted or score-
thresholded one therefore differs in two things at once:
a method emitting fewer instances, or maintaining fewer
identities, can post a better association score without asso-
ciating better. The benefit has to be measured at a *matched*
budget, against controls granted the same number of emit-
ted detections and, separately, the same number of emitted
identities. We study temporal confirmation as a streaming
emission policy, and ask first whether its consistency ben-
efit survives that comparison, then whether the benefit can
be retained without delaying the stream.

It survives. On nuScenes with a frozen CenterPoint de-
tector, confirmation improves association consistency un-
der both matchings, in both association frames, and against
confidence accumulation [1] as well as a static threshold,
so the gain is not an artifact of carrying fewer identities.
At the identical detection budget the same stream loses
detection recall, DetA and aggregate HOTA: the gain is
bought rather than added. Applied unchanged to an indoor
open-vocabulary system sharing no detector or modality,
the same rule reduces class-label switching by 26.9% at a
matched identity count.

Both costs descend from one decision, that an observa-
tion is emitted only once its track is confirmed. Releasing
the withheld observations after the fact delays them; dis-
carding them instead reverses the association gain outright
in the world frame. We therefore split the two decisions that
rule fuses: *when an output is emitted* and *when its status is*
confirmed. Every observation is emitted at the frame it oc-
curs, marked provisional; confirmation later changes only
its status, a shift in its track’s class vote relabels it, and a
track that never confirms is retracted within a bounded win-
dow. Confirmation leaves the critical streaming path: the

policy retains 160,938 observations (+43.6%) that prefix-deleting causal emission discards, at zero emission latency, and revision raises AMOTA by 0.0823. It is not free either: against that arm it gives up class-agnostic detection and identity quality, and a consumer that must hold each object the moment it comes close trades most of that arm’s misses for a far larger false-alert load (Sec. 5.5).

Contributions.

- **Streaming output policy.** We formulate streaming 3D perception as a provisional-output decision problem and develop a training-free policy that emits every observation at the frame it occurs and retrospectively confirms, relabels or retracts it as evidence accumulates, without emission-time latency; a bounded retraction horizon reproduces the unbounded retrospective output exactly.
- **Temporal-selection finding.** We show that temporal confirmation yields gains in outdoor identity consistency and indoor semantic consistency, and that these gains are not explained simply by reducing output volume.
- **Matched-budget evaluation.** We evaluate temporal confirmation under separately matched detection and identity budgets, and against a confidence-accumulation lifecycle, distinguishing temporal selection from changes in output volume and identity count.

2. Related Work

Streaming perception. Streaming perception scores predictions against the world at the moment they are *emitted*, folding latency into streaming AP (sAP) [16]; later work optimises detectors for this regime [35] and processes LiDAR sectors as they arrive rather than waiting for a full sweep [11]. sAP remains a per-frame average precision, so two systems emitting identical boxes at identical latency but differing entirely in identity continuity and label stability receive the same score.

Track lifecycle and confirmation. The classical radar literature decides when a candidate track is real enough to report with *track confirmation* — under M -out-of- N history logic a track is confirmed once it has been assigned a detection in M of the last N updates — and designs confirmation, deletion, and track management together [4]; the same literature established that reporting may lag estimation, *retro-diction* revising earlier estimates once later measurements arrive [15]. Our distinction is in the output semantics: we revise records already emitted to the streaming consumer, while retaining zero emission latency and allowing unconfirmed records to be retracted. Tracking-by-detection inherited the count-based half: SORT [3] and DeepSORT [33] withhold a tentative track until matched in N consecutive frames and AB3DMOT [32] carries the same minimum-hits rule onto 3D boxes, while CenterPoint [36] reports tracks from their first observation. We use this confirmation test

at its conventional operating point rather than proposing a new one.

Confidence in the track lifecycle. Recent 3D tracking has largely moved this decision from counting to confidence: CBMOT [1] replaces the minimum-hits criterion with a propagated tracklet score fused with each matched detection score, SimpleTrack [22] names *output* as a lifecycle decision separate from birth and death, and Poly-MOT [19] and Fast-Poly [20] pair category-specific association with mixed count- and confidence-based termination. We take this line as a comparator rather than a competitor: our associator is deliberately lightweight, so our absolute tracking scores are not comparable to theirs. Both rules decide which detections reach the output and both emit less, which is why we compare them at matched output and identity budgets.

Open-vocabulary 3D perception. This setting has moved from offline labelling of a completed reconstruction [5, 13, 23, 27] to incremental object maps [10, 12] and to fully on-line instance segmentation that maintains persistent identities across frames [28, 29, 34]. Cross-frame identity is thus an acknowledged design goal, but this work is scored by per-scene instance-segmentation AP, so how often a persistent instance’s *class label* changes as observations accumulate is never quantified.

Evaluation metrics. CLEAR MOTA [2] and IDF1 [24] weight detection and identity errors differently, while HOTA [21] factorises the two into DetA and AssA, which is what lets us read detection and association quality separately over an *identical* box set. In 3D, nuScenes AMOTA [6, 32] is class-aware and responds to labels the class-agnostic family ignores; video-consistency and open-vocabulary tracking metrics [9, 14, 17, 18, 31] instead require dense ground-truth tube identities over a fixed taxonomy, so we use existing metrics where computable.

Existing work targets streaming availability or instance-level consistency, but does not isolate temporal selection under matched output budgets, nor measure the class-label stability of persistent instances.

3. Method

3.1. Temporal lifecycle

Everything in this section is a post-processing operator over *frozen* detector output: no box coordinate or score is modified and no model is retrained, and the operator’s only freedoms are which detections enter the output stream, under which persistent identity, with which label, and at which frame. Per-frame detections — a 3D box with a score and a class label — are linked across frames into tracks by a nearest-centroid associator with a 2.0 m bird’s-eye-view **association threshold** and a **gap tolerance** of 5 frames; no score threshold is applied before association. Association runs either in the *sensor frame* (raw detector coordi-

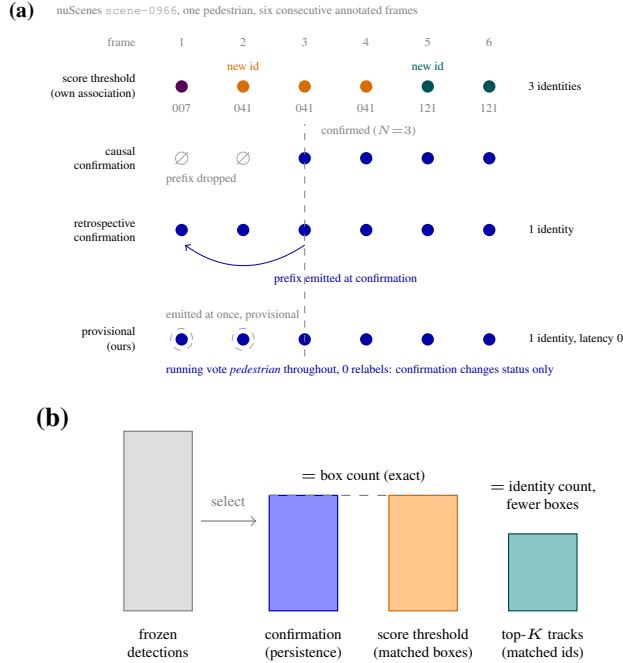


Figure 1. **Emission policies over the same frozen detections.** (a) One pedestrian over six annotated frames of nuScenes scene-0966; the emitted box is identical wherever it is emitted. Threshold identities (last three digits) come from the control’s own association pass; the other rows share one frozen association. Ours emits the first two observations at once as provisional (dashed ring) with the track’s running class vote, pedestrian throughout, so confirmation ($N=3$, third frame) changes their status, not their label. (b) The detection- and identity-budget controls match emitted boxes and emitted tracks respectively (Sec. 4.2).

nates), where a detection can continue only a track of its own class, or in the *world frame* (ego-motion-compensated global coordinates), where class is not consulted. Both are evaluated throughout, because ego-motion compensation changes how hard association is, and within each frame the associator is identical in every arm of every experiment. A track is **confirmed** once it has accumulated N observations; we use $N=3$ throughout, the conventional setting in tracking-by-detection systems, and do not tune it per dataset or per metric. Counting is *cumulative* over the scene rather than over consecutive frames, so a track seen at frames 0, 3, 6 confirms at frame 6. An unconfirmed track is **retracted** once $t - t_{\text{last seen}} > H$, where H is the *retraction horizon* in frames; confirmed tracks are never retracted, and $H=\infty$ applies no death rule and defers to a scene-end decision, which serves as the reference endpoint.

Throughout, the **confirmation module** denotes the standard composite that this test anchors: the confirmation test above, track-level majority-vote relabelling, which overwrites each emitted box’s class with its track’s voted class,

and track merging. All three are active in every arm labelled *Confirmation*; we write *confirmation test* when we mean the initialisation rule alone. The composite is conventional rather than an addition of ours, and holding it fixed is what lets our policy differ from it in emission only.

3.2. Emission policies

Over an identical lifecycle, three policies differ only in what the consumer sees and when (Fig. 1a). **Causal confirmation** emits nothing until confirmation and drops the pre-confirmation observations permanently. **Retrospective confirmation** emits nothing until confirmation, at which point the withheld prefix is released alongside the current observation, completing the trajectory at the price of emission latency. **Provisional emission with revision** (ours) emits every observation at its own frame and updates it as evidence accumulates.

3.3. Provisional emission and revision

Every observation is emitted at the frame it occurs under a *provisional status*: a flag on an emitted observation recording that the evidence supporting it is not yet complete. It is a property of an emitted record, not of a withheld one. Confirmation issues no prefix, since none was withheld; it changes only the status of the track and its observations. A *revision* is the module’s track-vote relabelling, read causally: when the running vote changes, before or after confirmation, it rewrites the label of the track’s emitted observations and their derived nuScenes attribute; geometry, score and identity are kept, so the emitted box set is unchanged. A track that reaches the retraction horizon unconfirmed is retracted.

Retrospective confirmation incurs $N-1$ observations of latency, and a larger empirical frame-index delay when observations contain gaps (Sec. 5.4); our policy has zero emission latency — asserted mechanically, since every emitted record carries the frame index of its own observation — while H bounds retraction rather than emission. What the policy requires instead is a consumer that can act on an update to a record it has already received.

4. Evaluation Protocol

For the controlled comparisons of temporal confirmation, we match two output budgets — emitted detections and emitted identities — against non-temporal selection rules rather than against the unfiltered baseline.

4.1. Datasets and pipelines

Outdoor. The nuScenes [7] validation split, all 150 scenes at the 2 Hz annotated-frame rate. Detections come from a pretrained CenterPoint [36] detector whose per-frame output is computed once and *frozen*; every arm of every experiment replays the same stored detections (718,786 boxes over the split; mAP 0.5613, NDS 0.6480).

Indoor. The ScanNet200 [25] validation split, all 312 scenes, in a streaming instance-segmentation pipeline: class-agnostic 3D instance-mask proposals from Mask3D [26] are labeled per frame by a 2D open-vocabulary detector [8], so each persistent instance receives a stream of per-frame class labels. The proposal set is likewise frozen across arms. This pipeline shares no detector, modality, or codebase beyond the temporal module with the outdoor one.

4.2. Matched controls

Two budgets are matched — emitted detections and emitted identities (Fig. 1b) — and under each we compare the module against a non-temporal score rule and against the other established temporal rule.

Detection-budget-matched control. A score threshold on the frozen baseline detections is calibrated so the control emits *exactly* as many boxes as the temporal module — 257,259 in the sensor frame (threshold 0.2214), 517,566 in the world frame (0.1272), both with zero count error. The match is exact because the matched threshold is the K -th largest observed score, ties resolved upward so the count never exceeds K .

Identity-budget-matched control. Per sequence, the control keeps the K tracks with the highest whole-track mean detection score, K being the number of identities the temporal module emits there; counts match exactly (41,207 tracks sensor, 68,222 world). Five additional arms select the same K tracks *at random* (five seeds), scored on association accuracy, separating score-based selection from selection per se. Every identity-budget arm here uses a per-sequence budget known only once the sequence ends, and the score-ranked ones whole-track scores: they are budget-matched references, not streaming policies, and comparisons against them hold emitted identities fixed. The two budgets cannot hold at once: at equal track count the control emits fewer boxes than the module (Tab. 2).

Confidence-accumulation control. To compare persistence against confidence rather than against a static score, a third control propagates a track score by constant decay and fuses it with each matched detection score, following CBMOT’s reported best update rule [1]; its parameters are transcribed from the authors’ released implementation, not tuned here, except that the gap tolerance is our associator’s rather than theirs, since the associator configuration is shared by every arm. At the detection budget a single threshold on the refined score is calibrated by the rule above; at the identity budget the control keeps, per sequence, the K tracks with the highest whole-track peak refined score, so it is matched under both budgets. Its trained score-regression variant is not reproduced: it is fit on nuScenes and would give the control supervision no other arm receives.

Indoor control. The identity-budget design transfers directly: the control keeps the K highest-scoring instances per scene, where K matches the temporal module’s emitted-instance count (10,413 instances in total; the control realizes 10,410). Three additional arms select K instances *at random* per scene (three seeds), to separate the effect of score-based selection from that of selection per se.

4.3. Metrics

Outdoor. Official nuScenes mAP and NDS on the emitted box set; class-agnostic HOTA, AssA, DetA, detection recall (DetRe), and IDF1 via TrackEval [21, 24], with similarity $\text{sim} = \text{clip}(1 - d_{xy}/2.0, 0, 1)$ matching the associator’s 2.0 m threshold; and official class-aware AMOTA [6] over the 7 nuScenes tracking classes; the two families are reported side by side.

Indoor. The primary endpoint is the per-scene count of *label switches*: frames in which a persistent instance’s emitted class label differs from its label in the previous frame *in which it was emitted*. Counting against the last emission rather than the last frame is what makes the count comparable across arms, since the arms emit on different frames. Prediction stability has precedent as a separate axis in video and 3D detection [30, 37], but that family scores box trajectories against ground-truth tracks, which ScanNet200 does not provide, and none of it scores the *class* assigned to a persistent instance. The secondary endpoint is instance-segmentation AP on the final reconstruction.

4.4. Statistics

For all three main comparisons — detection-budget-matched, identity-budget-matched, and indoor — the arms, the primary endpoint, and the criterion for a negative result were fixed before the run in a pre-registration released with the code. For the outdoor comparisons that criterion was that the association gain hold in *both* association frames with scene-bootstrap intervals excluding zero, and that the dataset-level metrics (mAP, AMOTA) move consistently across the two association frames, whichever direction they move in; indoors, that the per-scene label-switch count fall. The causal-emission ablation (Sec. 5.4) predates that protocol and is what motivated it. All comparisons between the temporal module and a control are paired per scene. We report the per-scene Wilcoxon signed-rank test with the rank-biserial correlation as effect size, and a 95% bootstrap confidence interval on the difference of the aggregate metric obtained by resampling scenes (10,000 resamples, fixed seed 20260718). A difference is described as an improvement only when its interval excludes zero; intervals that span zero are reported as no detectable difference, whatever their sign.

Intervals are available for the class-agnostic metrics, which are defined per scene and aggregate linearly over

scenes. AMOTA, mAP and NDS are not: all three are estimated over the whole split, so a per-scene resample does not yield the same estimator. We report them as point estimates and claim no interval support anywhere; the evidence we offer for AMOTA is that its sign and rough magnitude are stable across the four detection-budget-matched cells — retrospective and causal emission, each in both association frames — against both controls, and on a second detector (Sec. I). Matching emitted identities does not equalise the identities reaching the class-aware evaluation, which scores a subset of classes, so we report AMOTA only under detection matching.

5. Experiments

We report three outdoor comparisons on nuScenes val and one indoor comparison on ScanNet200 val, each between the confirmation-based module and a control granted the same output budget over shared frozen detections, and measure what the provisional stream is worth to a latency-sensitive consumer (Sec. 5.5). Unless stated otherwise, outdoor results use retrospective emission, whose latency cost is $N-1$ observations; the corresponding frame-index delay is reported in Sec. 5.4.

5.1. Association improves at a matched detection budget

Table 1 compares confirmation against controls calibrated to the identical box count — 257,259 boxes in the sensor frame and 517,566 in the world frame, from the same frozen detector output.

Association accuracy improves. Against the threshold control AssA rises from 0.2119 to 0.2744 in the sensor frame and from 0.5146 to 0.5293 in the world frame, both intervals excluding zero ($[+0.0471, +0.0785]$ and $[+0.0098, +0.0203]$). The per-scene mean gain, a different estimator from the difference of aggregates, is $+0.0460$; the gain holds in 113 of 150 scenes (Wilcoxon $p=2.6 \times 10^{-15}$, rank-biserial 0.74).

Class-aware tracking accuracy rises. AMOTA goes from 0.0942 to 0.1338 (sensor) and 0.3137 to 0.3864 (world), the largest relative effect we measure. AMOTA admits no per-scene resampling (Sec. 4.4); this is a point estimate.

Detection-oriented metrics fall. At the same budget, mAP drops $0.5504 \rightarrow 0.3009$ (sensor) and $0.5598 \rightarrow 0.4921$ (world), DetA drops $0.3570 \rightarrow 0.2232$ ($[-0.1408, -0.1267]$) and $0.2133 \rightarrow 0.1973$ ($[-0.0183, -0.0138]$), and detection recall drops $0.6160 \rightarrow 0.4287$ ($[-0.2087, -0.1665]$) and $0.6550 \rightarrow 0.6141$ ($[-0.0477, -0.0350]$). The DetA loss dominates the geometric mean, so aggregate HOTA falls as well ($[-0.0323, -0.0231]$ sensor; $[-0.0100, -0.0063]$ world).

IDF1 shows no reliable sensor-frame difference. The sensor-frame interval spans zero ($[-0.0108, +0.0037]$)

while the paired per-scene test reaches significance ($p=7.3 \times 10^{-3}$), so the two estimators disagree and we claim no difference; in the world frame IDF1 is slightly *lower* for the confirmation module ($[-0.0135, -0.0070]$).

Against confidence accumulation, the two rules split the metrics. This control isolates *which* temporal rule is used rather than whether one is used at all. Confirmation attains higher AssA in both frames ($0.2315 \rightarrow 0.2744$ sensor, $0.5110 \rightarrow 0.5293$ world; intervals $[+0.0337, +0.0524]$ and $[+0.0146, +0.0225]$), and accumulation is ahead on every other measured axis, also with intervals excluding zero: HOTA, DetA, IDF1 and DetRe (Sec. G). The split extends to detection quality: accumulation leads on mAP ($0.4783 \rightarrow 0.3009$ sensor, $0.5568 \rightarrow 0.4921$ world) and NDS ($0.6134 \rightarrow 0.5229$, $0.6467 \rightarrow 0.6200$).

Scope of the accumulation control. All statements about it concern our reproduction under the configuration of Sec. 4.2, not the original system or its published results (Sec. G).

5.2. The gain survives at a matched identity budget

Matching emitted identities rather than boxes (Tab. 2) tests whether the association gain merely reflects having fewer identities to confuse. It does not. AssA remains higher for the confirmation module in both frames ($0.2607 \rightarrow 0.2744$ sensor, $0.5074 \rightarrow 0.5293$ world), with intervals excluding zero ($[+0.0071, +0.0216]$ and $[+0.0177, +0.0267]$) and per-scene wins in 107/150 and 116/150 scenes ($p=3.9 \times 10^{-15}$ and 2.9×10^{-16}). Selecting the same number of tracks *at random* rather than by score does not reproduce the world-frame result: the seeded arms reach at most 0.4776, well below the confirmation module’s 0.5293. In the sensor frame the random arms are *not* separated from it — they span 0.2595–0.2765 and straddle the confirmation module’s 0.2744 — so the sensor-frame identity result rests on the score-ranked control alone.

The rule separation survives identity matching. Repeating the comparison against confidence accumulation with *identities* rather than boxes held fixed — the accumulation control ranked by its own whole-track peak refined score and cut at the confirmation module’s per-scene identity count, matched exactly — leaves the association separation unchanged. The paired per-scene mean Δ AssA is $+0.0327$ sensor $[+0.0262, +0.0397]$ and $+0.0244$ world $[+0.0189, +0.0301]$, both intervals excluding zero. The same per-scene mean estimator at matched boxes gives $+0.0341$ $[+0.0273, +0.0414]$ and $+0.0251$ $[+0.0196, +0.0310]$, with the accompanying losses on HOTA, DetA, IDF1 and DetRe keeping their sign and significance. The separation is similar in magnitude under either matched-budget constraint, which excludes the simplest reading of it — that persistence gains association accuracy by carrying fewer identities. The result is robust to

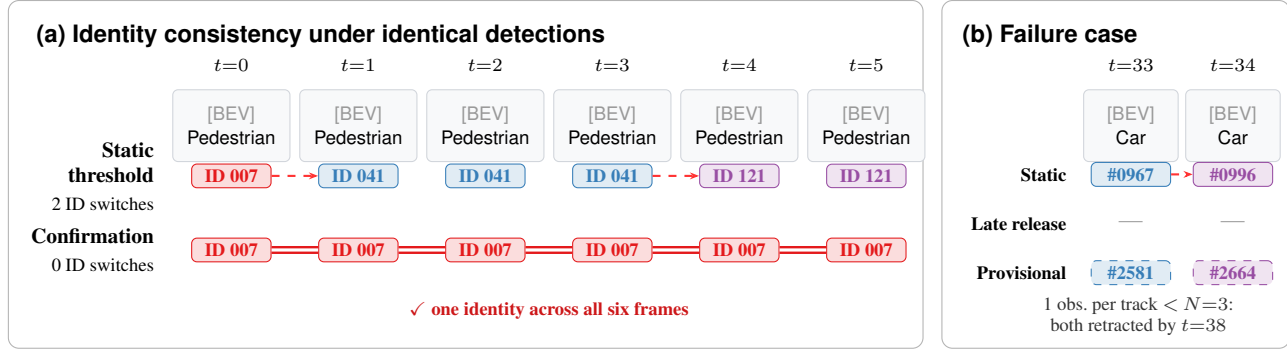


Figure 2. **Identity consistency under identical detections** (nuScenes *val*, sensor frame, detection-budget-matched arms; dashed red arrows mark identity switches; threshold identities come from the control’s own association pass). **(a)** The pedestrian of Fig. 1a (scene-0966). The emitted 3D box is identical in both arms in every frame — same centre, size, orientation and score (0.588–0.869, centre error ≤ 0.17 m, correctly classified throughout) — so the rows differ only in the identity attached to it. The threshold assigns three identities; confirmation with retrospective emission assigns one. **(b)** The cost: a correctly classified, well-localised car in scene-0104, annotated in two keyframes and held by the frozen association as two one-observation tracks, two short of $N=3$. The threshold emits it under two identities; late release never emits it; provisional emission emits both observations and retracts both tracks by $t=38$.

Frame	Arm	Boxes	mAP	NDS	HOTA	AssA	DetA	DetRe	IDF1	AMOTA
Sensor	Baseline (unfiltered)	718,786	0.5613	0.6480	0.1793	0.2007	0.1620	0.6690	0.1112	0.0858
	Control (threshold)	257,259	0.5504	0.6450	0.2735	0.2119	0.3570	0.6160	0.2082	0.0942
	Control (accumulation)	257,259	0.4783	0.6134	0.2604	0.2315	0.2954	0.5350	0.2146	0.1101
	Confirmation ($N=3$)	257,259	0.3009	0.5229	0.2462	0.2744	0.2232	0.4287	0.2054	0.1338
World	Baseline (unfiltered)	718,786	0.5613	0.6480	0.2837	0.5014	0.1613	0.6663	0.2715	0.2896
	Control (threshold)	517,566	0.5598	0.6479	0.3305	0.5146	0.2133	0.6550	0.3391	0.3137
	Control (accumulation)	517,565	0.5568	0.6467	0.3282	0.5110	0.2118	0.6511	0.3399	0.2931
	Confirmation ($N=3$)	517,566	0.4921	0.6200	0.3224	0.5293	0.1973	0.6141	0.3288	0.3864

Table 1. **Detection-budget-matched comparison** on nuScenes *val* (150 scenes, frozen CenterPoint detections, retrospective emission). The three matched arms — a static score threshold (0.2214 sensor, 0.1272 world), a confidence-accumulation control in the manner of [1], and the confirmation module (confirmation test, relabelling and merging; Sec. 3.1) — are each calibrated to the module’s emitted-box budget: the threshold control matches it exactly in both frames, the accumulation control exactly in the sensor frame and to within one box in the world frame. Bold marks the best matched arm, omitted where the leading interval spans zero; the unfiltered baseline is reference only and is not budget-matched.

the control’s ranking and matching granularity (Sec. B).

Budget asymmetry. Identity matching necessarily leaves unequal box counts, because score-ranked tracks are shorter than persistence-selected ones. This biases volume-sensitive metrics against the control — so the AssA result here is conservative — and we therefore read DetA, HOTA and IDF1 from Tab. 1, where boxes are matched exactly.

5.3. Confirmation-window sensitivity

All results so far fix $N=3$. Table S1 (supplementary) varies it over $\{2, 3, 4, 5\}$, each against its own exactly matched control ($N=1$ is excluded: it reproduces the unfiltered baseline detection-for-detection, so every difference is zero by construction). The association gain is positive at every N in both frames and all eight intervals exclude zero. It grows monotonically with N , close to linearly in the sensor frame and with visible deceleration in the world frame, while AMOTA flattens in the sensor frame and peaks at $N=4$ in

the world frame, and the detection cost rises without turning over at any N we measured.

No single N is best on every metric, and $N=3$ is best on none of them. We report this rather than re-select: tuning N on the test metrics is the practice the matched controls exist to guard against, and N is a latency budget — the $N-1$ observations of delay a deployment can absorb — not a hyperparameter. What the sweep establishes is that the consistency gain is a property of confirmation over the whole range tested, and that larger windows buy more of it at proportionally more detection cost and latency.

5.4. Policy behaviour and its trade-off

Sections 5.1 and 5.2 price confirmation in detection retention. This section asks what *else* it pays. All results so far release a confirmed track’s withheld prefix retrospectively; we first measure the causal alternative, then a policy that avoids both costs. Two things change here: these arms are

Arm	Boxes	HOTA	AssA	DetA	IDF1
<i>Sensor frame</i>					
Control (top- K)	138,053	0.2959	0.2607	0.3383	0.2542
Control (rand.)	—	—	.2595–.2765	—	—
Confirmation	257,259	0.2462	0.2744	0.2232	0.2054
<i>World frame</i>					
Control (top- K)	412,203	0.3630	0.5074	0.2610	0.3969
Control (rand.)	—	—	.4705–.4776	—	—
Confirmation	517,566	0.3224	0.5293	0.1973	0.3288

Table 2. **Identity-budget-matched comparison.** The control keeps, per scene, the K tracks with the highest whole-track mean score, a budget-matched reference rather than a streaming policy, K being the number of identities the confirmation module emits (41,207 sensor, 68,222 world; exact in both frames); the random- K rows (“rand.”) give the range over five seeds of the same selection made at random, scored on AssA only. The box budgets are unequal here, in the control’s disfavour on volume-sensitive metrics. AMOTA is omitted: identity matching does not equalise class-aware evaluation volume (Sec. 4.4).

Arm	Boxes	mAP	HOTA	AssA	AMOTA
<i>Sensor frame</i>					
Control (threshold)	168,193	0.5287	0.3044	0.2147	0.0997
Confirmation, causal	168,193	0.2047	0.2484	0.3088	0.1290
<i>World frame</i>					
Control (threshold)	361,426	0.5562	0.3864	0.5308	0.3417
Confirmation, causal	361,426	0.4480	0.3451	0.5184	0.4050

Table 3. **Causal emission is not free either: detection-budget-matched** (168,193 boxes sensor at threshold 0.3320, 361,426 world at 0.1673; both arms exact). The causal arm drops the pre-confirmation prefix instead of releasing it retrospectively. Intervals are reported for AssA only, the metric the comparison turns on; the remaining columns are point estimates.

not budget-matched, since the emitted box count is what an emission policy acts on, and the comparator is the confirmation test alone, without the module’s relabelling and merging.

Causal emission loses the prefix, and with it the world-frame gain. Table 3 repeats the detection-budget-matched comparison with the prefix permanently dropped. The result splits by association frame. The AMOTA gain survives in both (0.0997 $\rightarrow$ 0.1290 sensor, 0.3417 $\rightarrow$ 0.4050 world), and in the sensor frame the association gain is not merely preserved but larger than under retrospective emission (0.2147 $\rightarrow$ 0.3088; interval [+0.0722, +0.1162] against [+0.0471, +0.0785] retrospectively, 112 of 150 scenes). In the world frame, however, causal emission *loses* association accuracy relative to the control (0.5308 $\rightarrow$ 0.5184), with a bootstrap interval entirely below zero ([−0.0195, −0.0049]; paired $p=0.036$, rank-biserial −0.20, confirmation ahead in only 59 of 150 scenes). The detection cost is larger throughout than under retrospective emission (mAP 0.5287 $\rightarrow$

0.2047 sensor).

Causal emission is worse where association was already easy: ego-motion compensation leaves the world-frame control associating well on its own, and there the discarded prefix costs more than confirmation cleans up.

How much latency, in which unit. The $N - 1$ figure routinely quoted for retrospective emission is correct only in observations (Sec. 3.3): over the 150 scenes the confirmation *frame* horizon has median 3 and 95th percentile 7 — 1.5 s and 3.5 s at the 2 Hz annotated-frame rate, against a constant 2-observation horizon. It is unbounded in principle (Sec. C).

Provisional emission retains the prefix at zero latency. Table 4 measures the proposed policy against causal emission under the $N=3$ confirmation test alone — the prefix-deleting comparator, which applies neither the relabelling nor the merging of the confirmation module (Sec. 3.1) — over the identical frozen stream. At a retraction horizon of $H=4$ frames the policy emits 529,662 boxes against that arm’s 368,724: it retains 160,938 observations (+43.6%) that the arm deletes, and it does so at an emission latency of exactly zero frames. Detection quality on the emitted set rises accordingly (mAP 0.4483 $\rightarrow$ 0.4927, NDS 0.6013 $\rightarrow$ 0.6206) and so does association consistency (AssA 0.5265 $\rightarrow$ 0.5378); that same final volume selected by score alone (529,662 boxes, own association) reaches only 0.5132 (+0.0246, [+0.0198, +0.0300]), so this final-state figure is not a volume effect.

And it is not free. Over the same comparison the policy gives up class-agnostic detection and identity quality: DetA 0.2275 $\rightarrow$ 0.1936, aggregate HOTA 0.3452 $\rightarrow$ 0.3219, IDF1 0.3662 $\rightarrow$ 0.3305, and AMOTA before revision falls 0.4051 $\rightarrow$ 0.3823. Emitting a provisional record means emitting records that the prefix-deleting comparator would have suppressed and that ground truth does not support, and the class-agnostic metrics price exactly that.

Revision recovers the class-aware endpoint. Applying the completed-evidence label to the already-emitted stream takes AMOTA 0.3823 $\rightarrow$ 0.4646 at $H=4$ (+0.0823), leaving the net effect against the prefix-deleting comparator at +0.0594: revision restores, and then exceeds, an AMOTA the emission step had lost rather than adding one on top of it. It says nothing about association — the class-agnostic matcher does not read the class label, so those columns are a mechanical identity, not a null result.

The retraction horizon saturates. H prices how long a provisional record may still be withdrawn. At $H=4$ the emitted output becomes identical — byte-identical, verified by checksum — to the unbounded $H=\infty$ policy that defers the decision to scene end, because no track that ever confirms has a pre-confirmation gap wider than four frames. Every endpoint at $H \geq 4$ therefore coincides by output equality rather than by a tuned metric match. The bound re-

Arm	Boxes	mAP	NDS	HOTA	AssA	DetA	IDF1	AMOTA
Baseline (ungated)	718,786	0.5613	0.6480	0.2837	0.5014	0.1613	0.2715	0.2896
Causal emission, confirmation test only	368,724	0.4483	0.6013	0.3452	0.5265	0.2275	0.3662	0.4051
<i>Ours, provisional emission, retraction horizon $H=0$</i>								
provisional	295,403	0.3832	0.5651	0.3675	0.5619	0.2417	0.4198	0.3603
+ revision	295,403	0.3950	0.5701	—	—	—	—	0.4200
<i>Ours, provisional emission, retraction horizon $H=4$</i>								
provisional	529,662	0.4927	0.6206	0.3219	0.5378	0.1936	0.3305	0.3823
+ revision	529,662	0.5025	0.6247	—	—	—	—	0.4646

Table 4. **Provisional emission and revision**, world frame, nuScenes val (150 scenes, $N=3$, same frozen detections and association as Tab. 1). These arms are **not budget-matched**: the box count is what the emission policy acts on. Revision changes no box, score, or identity, so the class-agnostic metrics are *identical* by construction (“—”). The comparator is causal (prefix-deleting) emission under the confirmation test alone: unlike every arm labelled *Confirmation* elsewhere in this paper, it applies neither the track-vote relabelling nor the merging of the confirmation module (Sec. 3.1). Bold marks the best value among the policy arms and that comparator.

Arm	Miss rate ↓		False alerts ↓	
	Sensor	World	Sensor	World
Causal emission, confirmation test only	69.9%	47.9%	533	884
Score-ranked control, whole-track	14.4%	2.5%	542	1,036
Ours, provisional, $H=4$	2.6%	2.2%	2,098	1,545

Table 5. **What zero latency buys a consumer**, nuScenes val (150 scenes, $N=3$). Miss rate: fraction of objects the consumer does not hold at the frame they first come within 20 m of the ego vehicle, mean over scenes. False alerts: held records within 20 m that match no object, per 100 object-frames within 20 m. The first and last rows are the arms of Tab. 4 (and their sensor-frame counterparts); revision changes no entry. The consumer applies every record as it arrives and ignores the provisional status. The score-ranked control, matched to the same identity count, is a reference rather than a streaming policy: it ranks tracks by whole-track peak score, which is not available when a track is first emitted.

places the scene-end decision at no cost in output (Sec. A). **The horizon is itself a trade-off.** Tightening it to $H=0$ — retract at the first missed frame — reverses the picture: the policy then emits fewer boxes than the prefix-deleting comparator (295,403) and is the *best* arm in the table on every class-agnostic metric (AssA 0.5619, HOTA 0.3675, DetA 0.2417, IDF1 0.4198), while giving up mAP and AMOTA. We report both endpoints rather than selecting the one that flatters a chosen column: H buys retention and detection quality with class-agnostic identity quality, and $H=4$ is the setting at which the unbounded-horizon result is recovered in full at bounded latency.

5.5. What zero latency buys a consumer

Zero emission latency is a property of the stream; whether it is worth anything depends on what reads it. We measure it with the simplest consumer for which timing is the whole question: an object memory that must hold each ground-truth object at the frame the object first comes

within $R=20$ m of the ego vehicle. The memory applies every emitted record, revision and retraction as it arrives; an object is *missed* if no held record is assigned to it at that frame under one-to-one matching at the associator’s 2.0 m threshold, regardless of class, and a held record within R assigned to no object is a *false alert*. The 5,487 critical objects are defined from ground truth and ego pose alone, before any arm is read, and R lies inside the smallest official per-class evaluation range (30 m), so no class is truncated.

The prefix-deleting arm is still waiting when it matters. Against the causal comparator the policy cuts the miss rate by 67.4 points in the sensor frame ($[+64.1, +70.6]$), fewer misses in all 149 scenes that contain a critical object) and 45.7 points in the world frame ($[+41.6, +49.9]$, 147 of 149; Tab. 5). These are not late arrivals: 1,725 of the comparator’s 3,262 sensor-frame misses are never held before the object leaves the 20 m radius. The reduction keeps its sign with objects visible from a scene’s first frame excluded, at a 1.0 m matching distance, and at $R=10$ and 30 m (Tab. S2).

The consumer pays for it. The same memory holds far more records that correspond to nothing: about 95 additional false-alert frames per recovered object in the sensor frame (60 in the world frame). Against the score-ranked control the policy misses fewer objects in the sensor frame ($[+9.7, +14.1]$ points) at nearly four times its false alerts, and in the world frame the two miss rates show no detectable difference ($[-0.02, +0.67]$ points, $p=0.078$) while the control raises fewer false alerts. The advantage measured here is over waiting for confirmation; superiority over immediate emission is not established. A consumer that instead admits only confirmed records holds exactly the comparator’s memory, so the two arms are the endpoints of consumer policy rather than two points on a curve. Waiting cost is converted into correction cost, not removed. Revision’s share of the correction is narrow: in the world frame it cuts label reversals on critical objects from 1,296 to 887, and it changes no entry of Tab. 5.

Arm	Inst.	Switches	AP	AP ₅₀
Baseline (unfiltered)	—	—	0.1956	—
Control (top- K)	10,410	23,303	0.1926	0.2504
Control (random K)	—	23,275–23,288	—	—
Confirmation ($N=3$)	10,413	17,045	0.1954	0.2532

Table 6. **Indoor comparison** on ScanNet200 val (312 scenes, frozen Mask3D proposals with per-frame open-vocabulary labels). Primary endpoint: the per-scene label-switch count (lower is better, bold). The control keeps the K highest-scoring instances per scene, K matched to the confirmation module’s emitted-instance count; the random- K row spans three seeds. AP is secondary and untested. The unfiltered baseline is a validity anchor rather than an arm, and only its AP was recorded.

5.6. Indoor: semantic stability

The indoor experiment asks whether the same rule improves *semantic* stability in the pipeline of Sec. 4.1, which shares nothing with the outdoor one. Against the identity-matched control, confirmation reduces label switches from 23,303 to 17,045, a 26.9% reduction at an essentially equal instance budget (10,413 vs. 10,410). It holds in 309 of 312 scenes (Wilcoxon $p=1.5 \times 10^{-52}$, rank-biserial 0.99; per-scene distribution in Sec. F). The three random- K arms are indistinguishable from the score-ranked control and far from the confirmation module, so the reduction is specific to temporal selection rather than to emitting fewer instances by any rule.

The secondary endpoint shows no material change: AP 0.1954 against 0.1956 unfiltered and 0.1926 for the control. Unlike outdoors, the indoor gain therefore does not come at a measurable detection-quality cost — the two pipelines differ in whether the withheld detections carry segmentation mass, and we report the difference in effect without claiming the outdoor trade-off is avoidable in general.

The waiting is separable from the gain. Provisional emission moves the window’s waiting cost after emission rather than removing it. Figure 4 sweeps both on these scenes and proposals: provisional emission here fixes each emitted label H observations later (*revision horizon*); unlike the outdoor retraction horizon, H does not govern retraction, which removes the same 104 observations at every H . Confirmation at $N=2 \dots 5$ pays 2.31 to 9.53 frames of mean initial latency, provisional emission pays none, and the two families coincide pairwise — $H=2$ reaches the 17,152 switches of $N=3$, $H=4$ the 13,502 of $N=5$ — by construction rather than by measurement, so this is not an additional consistency gain, only a change in when the cost falls. Revision is that cost: 7.98% of emitted observations are revised at $H=2$ and 11.29% at $H=4$, at unchanged AP (0.19607 across every confirmation and provisional arm). At the wide end $N=5$ spends 9.53 frames of waiting and never emits 169 instances, against immediate emission with

11.29% revisions and 104 retractions.

6. Discussion

Over an identical box set, association consistency improves and detection-oriented accuracy degrades: the gain is bought rather than added, and each further observation of delay buys more of it at proportionally more detection cost. Against confidence accumulation it is a split rather than an ordering — persistence takes association consistency, accumulation keeps the detection- and identity-oriented measures — so the choice is which axis a system is consumed along, not a ranking.

Confirmation must decide what to do with the observations that precede it, and the classical rule offers two answers: release them late, or drop them and lose the world-frame association gain with them. Neither suits a consumer that must act immediately on output whose reliability improves later. The policy separates *when output becomes available* from *when evidence becomes reliable*, converting the cost from latency or retention into revisability (Sec. 5.5): a record already held may be updated or withdrawn, and the retraction horizon bounds how long that obligation lasts. It is an operating-point change, not a new confirmation mechanism.

6.1. Limitations

The revision endpoint is single. AMOTA is the only class-aware metric available to us; the class-agnostic family cannot respond to a semantic-only change, and its invariance in Tab. 4 is mechanical rather than a null result. Whether revision affects identity consistency would need a class-aware association metric, which we do not have and do not claim. Nor do we separate the label change from the nuScenes attribute recomputation it entails, so the AMOTA gain is attributed to the pair.

The association gain does not transfer across detectors. Repeating the budget-matched comparison on BEVFusion-L — LiDAR-only BEVFusion with a TransFusion head — reproduces the sensor-frame gain against all three controls (+0.0471, +0.0280, +0.0217 AssA, every interval excluding zero) but not the world-frame gain, which is negative against the threshold control (−0.0080, [−0.0144, −0.0015]) and indistinguishable from zero against both accumulation controls. The primary criterion of Sec. 4.4 therefore fails on this detector, and we narrow the association claim to the sensor frame; AMOTA stays positive in every cell, as a point estimate. The tested arm is confirmation, so this bounds Sec. 5.1 rather than the provisional policy; Sec. I reports the analysis.

Scope. The consumer of Sec. 5.5 is a proxy on one detector: it shows whether an object is held when it comes close, not what a planner does with it or whether the false-alert load is tolerable. ScanNet200 has no instance-tracking ground

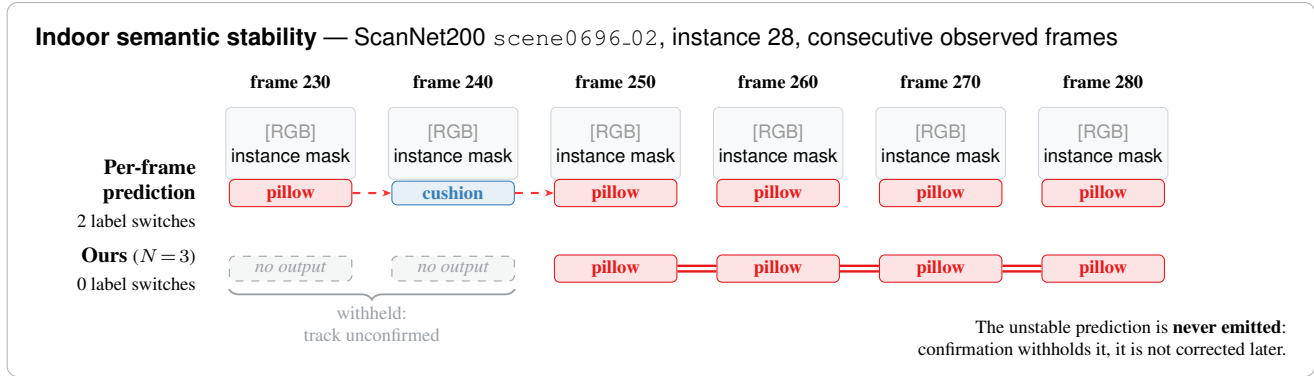


Figure 3. **Confirmation removes the unstable warm-up rather than correcting it.** One ScanNet200 instance (scene0696_02, instance 28) over its first six observed frames. The per-frame open-vocabulary prediction (top) disagrees with itself on the second observation, giving two label switches on a track that is otherwise a single class for all 50 observed frames. Our $N=3$ confirmation (bottom) emits nothing until the track is confirmed, so the disagreement never reaches the map; once confirmed, the label is held. Measured over all 17 candidate scenes, the confirmed output is identical to a suffix of the unconfirmed output for 602 of 604 instances, and *no* instance has its label changed after emission — the entire switch reduction comes from withholding the unconfirmed prefix, at a cost of two frames of latency here. Sequences are recorded per frame from a three-arm replay; the proposal cache was regenerated, so per-scene switch counts are not bit-identical to Tab. 6 (Sec. H).

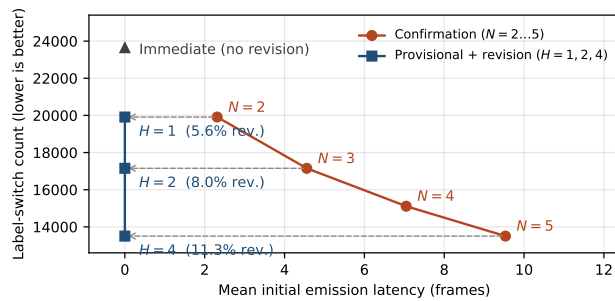


Figure 4. **Waiting versus revision** on ScanNet200 val. Confirmation ($N=2..5$) buys fewer label switches with pre-emission latency; provisional emission at revision horizon H reaches the same switch counts at zero latency, paying revisions instead (annotated). Arrows join the definitionally paired settings: the retrospectively finalised label at horizon H is the emitted label of the $(H+1)$ -th observation, so H and $N=H+1$ have identical switch counts by construction, not by measurement. All points come from one pass over the same frozen proposals, so counts differ slightly from Tab. 6; per-arm counts and the $H \rightarrow \infty$ bound are in Tab. S3.

output from evidence maturity: observations are emitted causally, then revised or retracted as evidence accumulates. On nuScenes, $H=4$ retains 43.6% more observations than the prefix-deleting comparator and revision raises AMOTA by 0.0823 over the provisional output. Against the prefix-deleting arm the policy gives up class-agnostic detection and identity quality. On ScanNet200 the same rule reduces class-label switching by 26.9% at a matched identity count. Confirmation is therefore best understood as an operating point in a detection–association–latency–revisability trade-off rather than a uniformly better tracker.

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truth, so the indoor endpoint is label switching rather than an identity metric and tests whether the confirmation rule transfers, not whether the streaming claim holds there.

7. Conclusion

Temporal confirmation provides a genuine association-consistency gain under matched detection and identity budgets, but the gain is coupled to detection retention and to latency, and on a second detector it survives only in the sensor frame. Provisional emission separates immediate

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818 A. Retraction horizon

819 At $H=4$ a track is confirmed or retracted within five frames
820 (2.5 s) of its last observation, instead of at scene end (up to
821 40 frames, 20 s). The bound costs nothing in output — the
822 emitted set is byte-identical to $H=\infty$ — and downstream
823 event traffic is unchanged by it (161.60 events per frame at
824 every H).

825 B. Control ranking robustness

826 Ranking the identity-budget control by mean instead of
827 peak refined score, or matching identities dataset-wide in-
828 stead of per scene, moves its AssA by less than 0.007 in
829 both association frames.

830 C. Confirmation latency distribution

831 The $N-1$ figure quoted for retrospective emission is a count
832 of *observations*. Over the 150 scenes the confirmation
833 *frame* horizon has mean 3.29, median 3, 95th percentile 7
834 and maximum 10; at the 2 Hz annotated-frame rate this is
835 a median of 1.5 s with a tail of 5.0 s, against a constant 2-
836 observation horizon. It is unbounded in principle.

837 D. Proximity-critical consumer: robustness

838 Numeric backing for the robustness statement of Sec. 5.5.

839 E. Waiting versus revision: per-arm counts

840 Full numeric backing for Fig. 4. Every arm is derived from
841 a single streaming pass over the 312 ScanNet200 val scenes
842 and one frozen proposal cache, verified byte-identical be-
843 fore and after the run; the confirmation rows are recomputed
844 from that pass rather than quoted from the separate run be-
845 hind Tab. 6, which is why their switch counts differ slightly
846 from it.

847 F. Indoor per-scene distribution

848 The per-scene label-switch reduction is 18.0 switches at
849 the median and 20.1 on average (95% CI on the mean
850 [18.7, 21.5]). The three random- K seeds of Tab. 6 give
851 23,275, 23,277 and 23,288 switches.

852 G. Confidence-accumulation control

853 At the matched detection budget the accumulation control
854 leads confirmation on HOTA ($[-0.0157, -0.0128]$ sensor,
855 $[-0.0065, -0.0052]$ world), DetA ($[-0.0805, -0.0645]$,
856 $[-0.0162, -0.0131]$), IDF1 ($[-0.0115, -0.0071]$,
857 $[-0.0123, -0.0099]$) and DetRe; all intervals exclude
858 zero. In the world frame the control is not statistically
859 separable from the static threshold on any of the five
860 metrics we resample (all intervals span zero); in the sensor
861 frame it is.

862 H. Indoor qualitative replay

863 The sequence in Fig. 3 comes from a three-arm per-frame
864 replay, not from the run behind Tab. 6: the frozen proposal
865 cache no longer exists and Mask3D regeneration is not bit-
866 reproducible, so per-scene switch counts differ from the ta-
867 ble. The mechanism the figure draws is exact — over the
868 replayed scenes the confirmed output is identical to a suf-
869 fix of the unconfirmed output for 602 of 604 instances, and
870 no instance has its label changed after emission — but the
871 frame indices are the replay’s own.

872 I. Cross-detector stress test: BEVFusion-L

873 Detector and protocol

874 The body’s tracking evaluation runs on one frozen Center-
875 Point detection set. To test whether the association gain of
876 Sec. 5.1 is a property of the confirmation rule or of that
877 detector, we repeat the same budget-matched comparison
878 on a second LiDAR detector: **BEVFusion-L**, LiDAR-only
879 BEVFusion with a TransFusion head, at $\text{mAP} = 0.6435$
880 and $\text{NDS} = 0.6918$ on nuScenes val. It is not a re-
881 parameterisation of the first: it is query-based and NMS-
882 free, emitting a fixed 200 queries per sample, where Cen-
883 terPoint is anchor-based with NMS in the loop. Proposal
884 density, score distribution and duplicate structure all differ,
885 which is what makes the swap informative.

886 Everything else is held fixed: nuScenes val, 150 scenes,
887 6,019 samples, $N=3$, retrospective emission, both associa-
888 tion frames, and the same matched controls. Every control
889 is a mask over one frozen box stream, so association, geom-
890 etry, scores and timestamps are shared by construction and
891 only the surviving set differs. The arm under test is *con-*
892 *firmation*, so this section bounds the premise of Sec. 5.1; it
893 does not evaluate the provisional policy of Sec. 3.3, which
894 we run on one detector only.

895 Budget-matched comparison

896 Tab. S4 repeats the detection-budget-matched compari-
897 son of Sec. 5.1 on this detector; Tab. S5 adds the two
898 confidence-accumulation controls. The three budgets are
899 met in different ways. The threshold control cuts a ranked
900 list of boxes and so matches exactly. The emission-budget
901 accumulation control admits whole tracks, so it can only
902 approach its budget from below and stops one box short in
903 each frame. The identity-budget control matches surviving
904 identities per scene exactly and leaves the box count free,
905 which is why its box count moves in opposite directions in
906 the two frames.

907 Transfer

908 Tab. S6 subtracts each control from the confirmation arm
909 and attaches the scene-bootstrap interval of Sec. 4.4. The

Frame	N	Delay	Boxes	Δ AssA	95% CI	Δ AMOTA	Δ DetRe
Sensor	2	1	375,982	+0.0296	[+0.0217, +0.0383]	+0.0176	-0.1210
	3	2	257,259	+0.0625	[+0.0471, +0.0785]	+0.0396	-0.1873
	4	3	203,019	+0.0933	[+0.0720, +0.1149]	+0.0478	-0.2217
	5	4	171,395	+0.1221	[+0.0960, +0.1480]	+0.0529	-0.2408
World	2	1	594,547	+0.0103	[+0.0068, +0.0140]	+0.0614	-0.0278
	3	2	517,566	+0.0148	[+0.0098, +0.0203]	+0.0726	-0.0409
	4	3	457,121	+0.0181	[+0.0120, +0.0250]	+0.0736	-0.0492
	5	4	409,463	+0.0199	[+0.0130, +0.0276]	+0.0682	-0.0556

Table S1. **Confirmation-window sensitivity.** Each row is confirmation at that N with retrospective emission, minus *its own* detection-budget-matched threshold control, matched exactly to the box count in the Boxes column. Delay is $N-1$ observations. The $N=3$ rows are the same comparison as Tab. 1, matched at the same budget. Δ AMOTA is a whole-split point estimate; the DetRe intervals are omitted for space and exclude zero at every N . The matched control’s own AssA rises with N as its budget tightens, from 0.2080 at $N=2$ to 0.2146 at $N=5$ in the sensor frame and from 0.5061 to 0.5260 in the world frame. This sweep is an exploratory sensitivity analysis.

Setting	Sensor frame		World frame		Scenes
	Δ miss	95% CI	Δ miss	95% CI	
$R=20$ m, 2.0 m match (main text)	+67.4	[+64.1, +70.6]	+45.7	[+41.6, +49.9]	149
First-frame objects excluded	+54.1	[+49.8, +58.5]	+25.8	[+21.6, +30.4]	145
1.0 m match	+67.9	[+64.7, +71.1]	+43.9	[+39.9, +47.9]	149
$R=10$ m	+57.2	[+53.1, +61.3]	+35.4	[+30.7, +40.3]	147
$R=30$ m	+70.4	[+67.6, +73.4]	+50.6	[+47.1, +54.1]	150

Table S2. **Robustness of the consumer result.** Each entry is the per-scene mean miss rate of causal emission under the confirmation test minus that of ours (provisional, $H=4$), in percentage points, with a 95% scene-bootstrap interval; Scenes counts those containing at least one critical object. Each row below the first changes one setting of the main-text comparison. “First-frame objects excluded” drops objects already visible at a scene’s first frame, which no confirming arm can hold at that frame. The critical set contains 2,656, 5,487 and 7,302 objects at $R=10$, 20 and 30 m.

Policy	Conf. horizon	Initial lat. ↓	LSC ↓	Rev. rate	Retract.	AP
Immediate	–	0	23,645	0%	0	0.19626
Confirmation	$N=2$	2.31	19,912	–	0	–
Provisional	$H=1 \equiv N=2$	0	19,912	5.56%	104	0.19607
Confirmation	$N=3$	4.56	17,152	–	0	0.19607
Provisional	$H=2 \equiv N=3$	0	17,152	7.98%	104	0.19607
Confirmation	$N=5$	9.53	13,502	–	0	–
Provisional	$H=4 \equiv N=5$	0	13,502	11.29%	104	0.19607
Provisional (bound)	$H=\infty$	0	0	20.94%	104	0.19607

Table S3. **Waiting versus revision** on ScanNet200 val (312 scenes, 448,690 emitted observations). Initial latency is the mean over emitted instances, in stream frames. LSC is the label-switch count; revision rate is revised observations over emitted observations. H is the revision horizon in observations: an emitted label is final H observations later. Retraction does not depend on H . Each provisional row is paired with the confirmation setting whose switch count it matches *by construction* ($H \equiv N-1$), not by measurement. AP is blank for $N=2$ and $N=5$: it was not computed in this evaluation, and we do not import it from another run. The grey row is the $H \rightarrow \infty$ retrospective bound — every observation carries the instance’s final label — which is a bound on what revision can recover, not an operating point.

sensor-frame rows reproduce the body’s result on a detector the body never saw; the world-frame rows do not.

Isolating emission from revision (both detectors)

The confirmation module includes track-voted relabelling after emission (Sec. 3.1); in the world frame it overwrites 244,943 emitted boxes. Its AMOTA therefore mixes the emission decision with a semantic revision, while the accumulation controls only ever change which boxes survive. We remove the confound with an arm that keeps the confirmation arm’s emission decision and restores the original per-box labels — same 858,279 boxes, same 117,023 identities, labels untouched.

Its class-agnostic metrics come back *bit-identical* to the confirmation arm (HOTA 0.2483, AssA 0.4869, DetA 0.1272, IDF1 0.2241). This is mechanical, not a null result: those metrics are computed by a class-agnostic matcher and cannot respond to a label change, the same invariance noted in Sec. 6. It also means the world-frame AssA result of Tab. S6 is unaffected by the relabelling.

AMOTA does respond: it falls from 0.3121 to 0.2793,

and AMOTP rises from 0.8781 to 0.9448. So 0.0328 of the confirmation arm’s world-frame AMOTA margin was the revision rather than the emission decision. Tab. S6 reports the emission-only figures.

Detection cost

The cost the body reports is larger on this detector. In the sensor frame confirmation takes mAP from 0.6435 to 0.3500 and NDS from 0.6918 to 0.5497, while the threshold control spending the identical box budget holds mAP at 0.6423 and NDS at 0.6918. In the world frame the same policy takes mAP to 0.5706 and NDS to 0.6610 against a control that gives up nothing. The association gain, where it exists at all, is bought at a detection price that a matched non-temporal rule does not pay.

Frame	Arm	Boxes	mAP	NDS	HOTA	AssA	AMOTA
Sensor	Baseline	1,203,751	0.6435	0.6918	0.1404	0.1964	0.0977
	Threshold	460,905	0.6423	0.6918	0.2222	0.2062	0.1149
	Confirmation	460,905	0.3500	0.5497	0.1979	0.2532	0.1306
World	Baseline	1,203,751	0.6435	0.6918	0.2175	0.4721	0.2077
	Threshold	858,279	0.6435	0.6918	0.2615	0.4949	0.2400
	Confirmation	858,279	0.5706	0.6610	0.2483	0.4869	0.3121

Table S4. **BEVFusion-L, detection-budget matched.** The threshold control emits exactly the confirmation arm’s box count in both frames. Bold marks the better of the two matched arms. In the world frame the control takes AssA, which it does not in Tab. 1.

Frame	Control	Boxes	Ids	HOTA	AssA	DetA	AMOTA
Sensor	Emission	460,904	80,952	0.2056	0.2253	0.1889	0.1205
	Identity	453,716	78,290	0.2050	0.2316	0.1825	0.1208
World	Emission	858,278	105,381	0.2541	0.4858	0.1334	0.2111
	Identity	894,559	117,023	0.2493	0.4843	0.1288	0.2106

Table S5. **Confidence-accumulation controls on BEVFusion-L.** The emission-budget control is track-granular — a track is emitted whole or not at all — so it never exceeds its budget and a small shortfall is reported rather than removed: 460,904 of 460,905 in the sensor frame and 858,278 of 858,279 in the world frame. The identity-budget control matches the confirmation arm’s surviving identity count exactly, per scene, in all 150 scenes (78,290 and 117,023); its box count is unconstrained by construction and drifts in both directions.

What this section does and does not establish

Δ AMOTA carries no confidence interval, for the reason given in Sec. 4.4: AMOTA integrates MOTA over recall thresholds estimated on the whole split and has no per-scene decomposition, so the scene-resample protocol does not yield the same estimator. Producing one would require re-running the official evaluation once per resample, which is a different protocol from the one fixed before the run; we did not do it. The AMOTA pattern here is therefore a replicated point estimate — positive in all six cells of Tab. S6 and in the CenterPoint cells of the body — and not a statistically bounded effect.

Two further limits. DetRe is not reported for this detector, so Tab. S4 is not directly comparable to Tab. 1 column for column. And this is one additional detector: it is enough to refute a detector-independent reading of the association gain, which is what we use it for, and not enough to characterise how the gain varies with detector design.

Frame	Control	Δ AssA [95% CI]		Δ AMOTA
Sensor	Threshold	+0.0471	[+0.0356, +0.0588]	+0.0156
	Emission	+0.0280	[+0.0220, +0.0340]	+0.0101
	Identity	+0.0217	[+0.0168, +0.0268]	+0.0097
World	Threshold	−0.0080	[−0.0144, −0.0015]	+0.0393
	Emission	+0.0011	[−0.0017, +0.0041]	+0.0682
	Identity	+0.0025	[−0.0003, +0.0056]	+0.0687

Table S6. **Confirmation minus control on BEVFusion-L.** Intervals are the scene-bootstrap protocol of Sec. 4.4, unchanged (10,000 resamples, seed 20260718). In the sensor frame all three intervals exclude zero. In the world frame the threshold interval excludes zero *below* it and both accumulation intervals span zero, so the world-frame effect is zero or negative under every control — the opposite of the world-frame result in Tab. 1 and Tab. S1. Δ AMOTA is a point estimate with no interval (Sec. 4.4); the world-frame values are emission-only, from the control of §E. We do not tabulate CenterPoint deltas beside these, because the CenterPoint accumulation arms were matched at a different budget than Tab. 1 and the two are not one comparison.